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Self-tuning fuzzy PID speed controller for quarter electric vehicle driven by In-wheel BLDC motor and Pacejka's tire model

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Abstract: Vehicles using internal combustion engine (ICE) adversely affect the environment. Therefore, all vehicle makers have started their mutation toward electric vehicles. In this paper, a quarter vehicle model is developed considering brushless DC (BLDC) motor actuation. The longitudinal behavior of the drive wheel is modeled using the Pacejka model or magic formula. This model describes the interaction between the tire and the road surface, which reflects road condition effect. The purpose of the control is to provide smooth speed control in both acceleration and deceleration modes, regardless of changes in road conditions. Due to the high complexity of the control model, a hybrid fuzzy-PID controller is designed to serve as a speed controller for the system motor-tire-road-surface. It is checked by numerous simulations that the proposed controller performs well, ensuring comfortable steering operation despite road characteristic changes.

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Keywords: Longitudinal vehicle dynamic, 2-DOF vehicle longitudinal model, longitudinal slip, tire Pacejka model, in-wheel motor BLDC, fuzzy logic control.

1. INTRODUCTION

Climate changes and atmospheric pollution, mostly caused by human actions, are major worldwide concern. To remediate this issue, greenhouse gas (GHG) reduction remains the only possible way (Meyer et al., 2012). As a matter of fact, transportation industry is responsible for substantial impact on atmospheric gas emissions (Ternel et al., 2021). Therefore, an interesting progress has been made, in recent years, in the development of electric vehicles and hybrid electric vehicle (HEV). In this respect, an intensive research activity has been devoted to topics like energy efficiency and drastic consumption reduction of fossil-based fuels (Un-Noor et al., 2017), suspension control (El Majdoub et al., 2020), longitudinal motion control (El Majdoub et al., 2012; El Majdoub et al., 2013). Obviously, electric vehicles (EVs) meet these requirements. Electric motors and static converters are among the main components of EVs (Cai et al., 2021). The wheels can be driven directly or indirectly through gearbox and differentials by an electric motor. But, In-wheel motors are most commonly used in direct drive vehicles..(De Novellis et al., 2015).

Indirectly controlled electric vehicles are propelled by the wheels or, simply, by wheel-motors, which are powered by electric motors through transmission and differential gears (El Majdoub et al., 2013). Directly driven EVs are propelled by in-wheel or, simply, wheel-motors. The main of this research is to develop the control of the In-wheel motors placed and operated within a pure EV. These wheel motors may be controlled independently and have such a quick and accurate reaction to control. Therefore, the vehicle chassis control or motion control becomes more stable and robust (He et al., 2005).

In-wheel motors are made by assembling electric motors and wheels in a way that avoids gears and chains, making useless mechanical power transmission parts, reducing thus energy losses.

The complexity of the resulting in-wheel based system depends on the way the road-tire contact is modeled. When simple models are used, system control can be performed using relatively simple control techniques, e.g. PI or PID controllers. The point is that such simpler approach, involving less accurate simple models and simple controllers, cannot provide quite satisfactory control performances, combining passenger comfort and vehicle security and reliability. To achieve high control performance, more accurate, which are generally more complex, models are needed to describe the in-wheel system including the tire-road contact. The point is that level of complexity of the resulting models is such that the available structural model-based control design methods, even nonlinear ones, prove to be inapplicable. Generally, the FLC approach features relative simplicity and satisfactory performances, including some robustness to modeling errors and others (Buckley, 2001).

In this paper, we are considering the problem of controlling the longitudinal movement of quarter vehicle directly driven by an in-wheel BLDC motor. The control objective is vehicle speed regulation, despite the changing road characteristics. The wheel is actuated on through an inverter, while rotor position measurements are provided by magnetic sensors placed in the rotor, defining the commutation of phases. The controlled system is modeled by applying electro-mechanical laws, to the corresponding parts of the system, and the Pacejka model (or

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magic formula) to the tire-road contact. The resulting model proves to be highly complex and cannot be used in control design applying the available (nonlinear) control methods. The traditional methods are simple to use, nevertheless it has several drawbacks, such as, that whenever the system state changes, the PID parameters must also change, resulting in decreased steady and dynamic behavior, poor reaction, and frequent oscillations. Moreover, it is not very robust to disturbances and noise. Therefore, a controller is designed that combines FLC and PID control. Presently, FLC is resorted to tune online the parameters of the PID control law, leading to a fuzzy adaptive PID speed controller. The FLC parameter tuning is based on two informations: the speed tracking error and error rate.

The paper is organized as follows: Section is devoted to modeling the controlled system including vehicle dynamics, in-wheel BLDC motor, and road-tire contact; the proposed fuzzy adaptive speed PID controller is designed in Section III; the performances of the proposed controller are highlighted in Section IV by simulation results performed with MATLAB/Simulink; a conclusion ends the paper in Section V.

2. LONGITUDINAL MOTION MODELLING OF THE VEHICLE

2.1 Driving wheel model

The drive wheel is an electromechanical system with a BLDC motor integrated into a tire. While there is a difference between an in-wheel motor and a standard wheel, by removing the wheel from the vehicle reveals all the principal elements. The area included is the braking system, an active suspension system, and the electric motor that drives the wheel. The most interesting point about in-wheel motors is

that the needed power is sent to the wheels directly. Because shortening the path between the motor and the tire, there is an enhancement of the effectiveness of the torque motor (Li et al., 2019). Figure 1 depicts the In-Wheel motor.

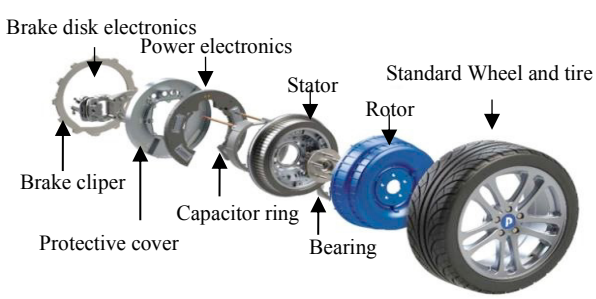

Figure 1: The component of the In-Wheel motor.

2.2 In-Wheel motor model

In this paper, the permanent magnet synchronous motor with trapezoidal back EMF (Electromagnetic Field) (BLDC) is adopted to motorize the driving wheel as an In-wheel motor in as much as its higher efficiency, power density, fast torque reaction, and large size speed range (Mishra et al., 2020). The BLDC motor is a type of DC motor where commutation is conducted electronically using hall sensors to detect rotor position. The three-phase of BLDC motor has more advantages in regards to the conventional motors such as high efficiency, high rated speed, torque/speed, torque during mechanical shocks, dynamic response, operation life time, less maintenance, noise susceptibility, and low cost (Shafiei et al., 2015). BLDC motors maybe with inner or outer rotor. Both types are built up with permanent magnets. The total of magnetic parts is equivalent to the number of poles of the motor. The windings of phases are disseminated on the stator with a trapezoidal electromagnetic field. The Phase to Phase back EMF and the structure of the BLDC motor are exposed in Figure 2. The BLDC motor equivalent circuit is illustrated by the following Figure 3.

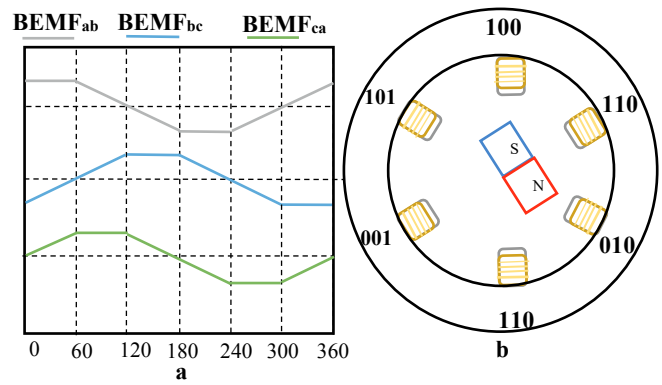

Figure 2: (**a**) Phase to Phase back EMF of BLDC motor. (**b**) Construction of BLDC motor

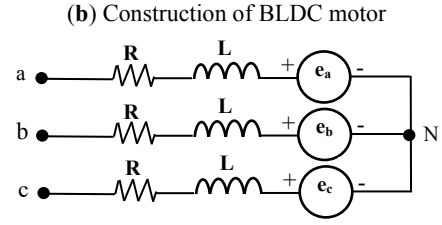

Figure 3: Equivalent circuit of BLDC motor

The matrix equations of three-phase voltages of BLDC motor (Ammari et al., 2021) is expressed as equation (1):

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (1)$$

Where v_a, v_b, v_c and i_a, i_b, i_c are three-phase voltages and currents, e_a, e_b, e_c are the phases back EMF, R, L and M are the stator resistance, the inductance and the mutual inductance respectively. In Equations 2 and 3, the electromagnetic and mechanical torque equations are shown (Al-Fiky et al., 2019).

$$T_e = \frac{i_a \cdot e_a + i_b \cdot e_b + i_c \cdot e_c}{\omega} \quad (2)$$

$$T_e = j_r \dot{\omega} + K_f \omega + T_L \quad (3)$$

where j_r, K_f, T_L and ω are, rotor inertia, friction constant, load torque and the rotor speed respectively.

2.3 Pacejka Tire Model

The tire is also one of the most challenging elements to be modeled: its physical constitution and behavior are very difficult to equate. However, it is essential to carry out a dynamic study of the tire since it is the part that ensures the link between the vehicle and the road. Furthermore, with the exception of aerodynamic forces, the wheel-road contact generates all disturbing forces applied to the vehicle, in this

respect several models have been proposed (El Majdoub et al., 2012). In this paper, the Pacejka tire model is retained to obtain the adhesion coefficient produced between the tire and the road surface as a function of longitudinal slip. The coefficient of adhesion μ is used to determine the longitudinal force F_x. This model is obtained by an empirical approach (Kogbara et al., 2016). This actively demonstrates that the coefficient of adhesion μ as a function of the slip ratio σ with the following empirical equation (4):

$$\mu(\sigma) = D\, sin(C tan^{-1}(B\sigma - E\,(B\sigma - tan(B\sigma)))) \quad (4)$$

where the parameters B is the stiffness factor, D is the shape factor, C is the peak value and E is the curvature factor (Bakker et al., 1987).

The values of B, D,$C$and E for various types of asphalt road conditions are provided in Table I:

Table 1. The coefficients used in the Pacejka equation according to the asphalt conditions.

	Dry	wet	snow	Ice
B	*12*	*12*	*5*	*4*
C	*1,9*	*2,3*	*2*	*2*
D	*1*	*0,82*	*0,3*	*0,1*
E	*0,97*	*1*	*0,1*	*1*

The longitudinal force F_x is calculated from the coefficient of adhesion and the normal force F_v applied to the wheel as shown in the following equation:

$$\mu = \frac{F_x}{F_v} \quad (5)$$

The slip ratio is different when the vehicle is in the acceleration or deceleration mode, it is defined as follows (Botha and Schalk Els, 2015):

$$\sigma = \frac{r\omega - v_x}{\max\,(v_x, r\omega)} \quad (6)$$

in the acceleration mode:

$$\sigma = 1 - \frac{v_x}{r_{eff}\,\omega} \quad (7)$$

in the deceleration mode:

$$\sigma = \frac{r_{eff}\,\omega}{v_x} - 1 \quad (8)$$

where r_{eff} is the wheel radius (m), v_x is the velocity of the vehicle (m/s) and ω is the rotational velocity of the wheel (rad/s).

2.4 Rolling resistance

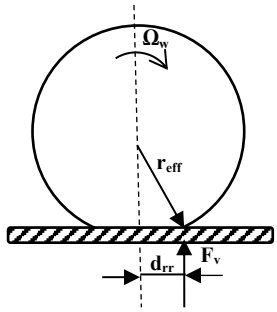

Figure 4: Rolling resistance

The rolling resistance is the compensating torque M_{rr} resulting from the forces that are responsible for moving from the central point to a distance $d_{rr,\max}$ which is defined as the maximal trail of vertical load.

$$d_{rr,max} = \mu_{rr} r_{eff} \quad (9)$$

$$M_{rr} = F_v d_{rr,max} = F_v \mu_{rr} r_{eff} \quad (10)$$

where μ_{rr} is the coefficient of rolling resistance and F_v the normal force.

2.5 vehicle quarter model

Let's consider the quarter vehicle shown in Figure 5. The effects of lateral and vertical forces are supposed ignored, and that the vehicle follows a linear trajectory. By applying Newton's first law which is based on the balance of forces acting on the vehicle, the acceleration/deceleration of the vehicle can be obtained as follows:

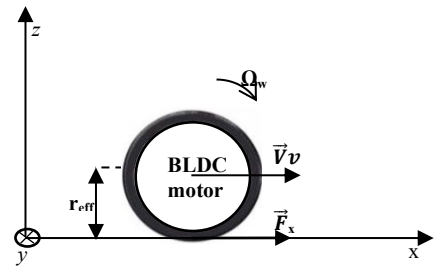

Figure 5: Quarter vehicle model

$$m\dot{v}_x = F_x \quad (11)$$

where m is the masse of the quarter vehicle (kg), $\dot{v}_x$ is the velocity of the vehicle (m/s) and the F_x is the traction force generated between the tire and the road surface acting on the wheel (N).

Applying the Newton's second principle, which correspond to the sum of torsional moments on a given axis is the product of the tensor of inertia J of the rotational acceleration along that axis, considering the rolling resistance; the equation will be established as follow:

$$T_e = J\dot{\Omega}_\omega + F_x r_{eff} + M_{rr} \quad (12)$$

where T_e is the electromagnetic torque produced by the In-Wheel motor (Nm), J is the moment of inertia of the driving wheel (kgm^2), Ω_ω is the rotational velocity of the driving wheel (rads^{-1}), r_{eff} is the wheel radius (m) and F_x is the traction or braking force (N).

3. The Design of the In-wheel BLDC motor Speed Controllers

3.1 Fuzzy PID controller

PID stands for "Proportional, Integral, and Derivative", as these are the tree actions it performs. A PID controller is a feedback control mechanism that calculates the deviation or the error between a detected value and the desired value according to apply a corrective action to adjust the process. The transfer function of PID controller is (S is the Laplace variable):

$$C(S) = K_p + \frac{K_i}{S} + K_d S \quad (13)$$

Where: K_p is the proportional gain, K_i is the integral gain, and K_d is the derivative gain. **Table 2** illustrates the impacts of each controller parameter, K_p, K_i, and K_d, on a closed-loop system.

Parameter increase	Rise time	Overshoot	Settling time	Steady state error

K_p	-	+	Small change	-
K_i	-	+	+	-
K_d	Small change	-	-	Small change

Table 2: PID control parameter characteristics

The linguistic terms of fuzzy logic are most of them expressed as logical intimations, such as 'If…Then' rules. These rules identify a variety of standards called fuzzy membership functions (Figure 6). Fuzzy membership functions can have a triangle shape, a trapezoidal shape, and others. The fuzzy controller computes an error value (e) as the distinction between the set speed and the measured speed. The proposed fuzzy logic controller has 2 inputs and 3 outputs (Fig. 7). Seven membership functions have been suggested, there are: NB, NM, NS, ZE, PS, PM and PB for each input and output. In Figure 7, The studied controller has been constructed to contain two inputs (error e and its derivative $\dot{e}$) and three outputs (K_p, K_i and K_d).

Tables 3–5 detail the rule base. There are seven language labels in the inputs e and $\dot{e}$,and seven linguistic labels in the outputs. After then, there will be a basis of 49 rules for each output. For the exact purpose of implementing the strategy of the fuzzy adaptive PID algorithm, the fuzzy relationship between three PID parameters and the deviation e and the rate of change of the deviation $\dot{e}$ must be established by continuous tests of e and $\dot{e}$, and then tune three PID parameters by the fuzzy reasoning method to meet the different requirements of the controller parameter with e and $\dot{e}$ changing. So that the controlled object has excellent dynamic and static performance. The block diagram of the fuzzy PID controller is illustrated in Figure 7.

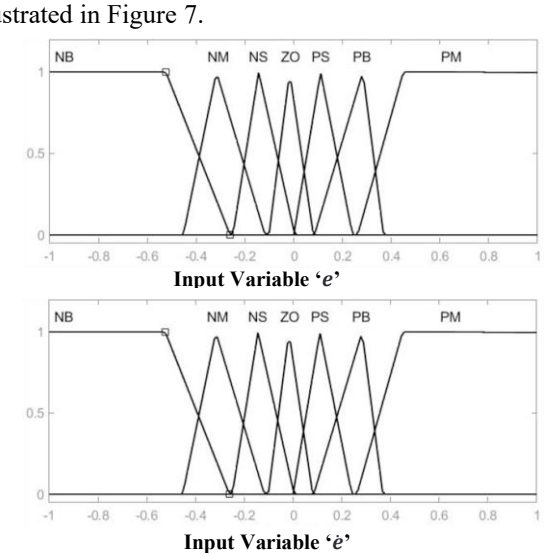

Figure 6: The member ship function of the input e and $\dot{e}$

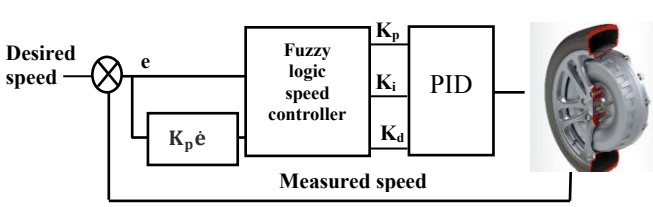

Figure 7: Diagram of a fuzzy PID controller coupled to an in-wheel BLDC motor.

The speed controller that is implemented to get good performance is outlined in this part. In fuzzy logic controller, knowledge base is composed of data and rule bases. Rule base contains fuzzy logic rules which are designed to achieve the purpose of control. Basic usage purpose of rule base is to be able to show expert information in manner of cause and effect. But we can create rules based on information and experiences of an expert, according to the setup of fuzzy model of the system or using various learning algorithms (Yildiz et al., 2013). Fuzzy rules base for each output K_p, K_i and K_d of the system is given in the tables 3, 4 and 5.

ė \ e	NB	NM	NS	ZO	PS	PM	PB
NB	VB	VB	VB	VB	VB	VB	VB
NM	MB	MB	M	M	M	S	MS
NS	B	B	B	S	S	MS	M
ZO	ZE	ZE	ZE	ZE	MS	MB	VB
PS	B	B	B	MS	MB	VB	VB
PM	MB	MB	MS	MS	VB	VB	VB
PB	VB	VB	VB	VB	VB	VB	VB

Table 3: The rule base of K_p

ė \ e	NB	NM	NS	ZO	PS	PM	PB
NB	M	B	MB	S	S	B	M
NM	M	B	MB	S	S	B	M
NS	M	B	MB	S	S	B	M
ZO	M	B	MB	ZE	S	B	M
PS	M	B	MB	MS	S	B	M
PM	M	B	MB	MS	S	B	M
PB	M	B	MB	MS	S	B	M

Table 4: The rule base of K_i

ė \ e	NB	NM	NS	ZO	PS	PM	PB
NB	ZE	VS	MS	S	V	VB	VB
NM	MS	MS	S	S	B	MB	MS
NS	MS	M	S	S	VB	VB	VB
ZO	M	MB	MB	VB	VS	MB	VB
PS	B	B	VB	VB	VB	VB	VB
PM	MB	VB	MS	VB	VB	VB	VB
PB	VB	VB	VB	VB	VB	VB	VB

Table 5: The rule base of K_d

4. SIMULATION

The model of the studied quarter vehicle is built on Matlab/Simulink (Fig 7), where the Model parameters used in this paper are presented in Table 7. We will evaluate the efficiency of the controller during acceleration and deceleration with a different type of roads (wet and dry), according to the Table 6. On the other hand, we will evaluate its robustness to the effects of the white noise that affects the signal measurement. The system is controlled by a filtered input close to the real driver's acceleration and deceleration command profile as shown in Figure 8. The system is exposed to a white noise which represents 5% of the measured speed value in other to evaluate the robustness of the controller.

The response of the system is acceptable, which follows the desired speed smoothly, see Figure 8, without an overshoot in acceleration of the vehicle, and the slip ratio developed by the wheel is less than *2%* on the dry road, but in the wet road, which represent a small difference between the slip that occurs between the tire and the road surface when driving on dry road, reaches *2.2%* as it is shown in the Figure 9.

However, with applying white noise at *20s* with range of 5%, the controller has a certain robustness to resist noise and the vehicle and the wheel speed are a slight variation as shown in Figure 8. The Figure 10 shows the variation of the longitudinal force produced between the road and the tire; it has a positive value during the acceleration by contrast of deceleration with a negative value.

Table 6. Simulated operation conditions

Time (s)	***0s -5s***	***5s -10s***	***10s -15s***	***15s -20s***	***20s -25s***
Speed (*km/h*)	*60-80*	*80-60*	*60-80*	*80*	*80*
Road state	*dry*	*dry*	*wet*	*dry*	*dry*
Noise (%)	*0*	*0*	*0*	*0*	*5*

Table 7: Model specification

g	Gravity acceleration	$9.81\ ms^{-2}$
J	Driving wheel inertia	$0.75\ kgm^2$
m	Mass of quarter of the vehicle	*270 kg*
r_{eff}	Wheel radius	*0.32 m*
μ_{rr}	coefficient of rolling resistance	*0.025*
Motor parameter:		
R	Stator resistance	*0.2 Ω*
L	Inductance	*0.85 mH*
j_r	Rotor inertia	$1kgm^2$

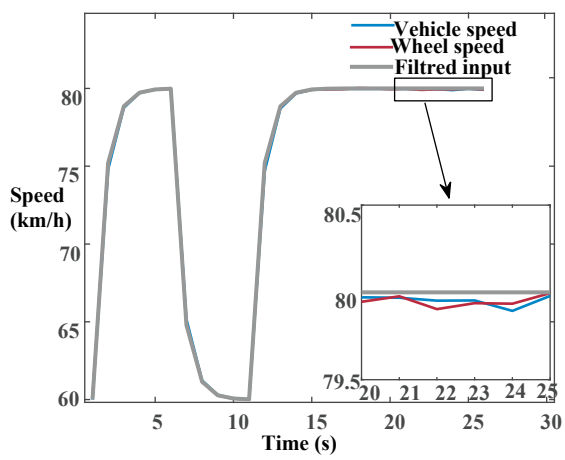

Figure 8: Vehicle and wheel speed and the noise effect on the controlled speed

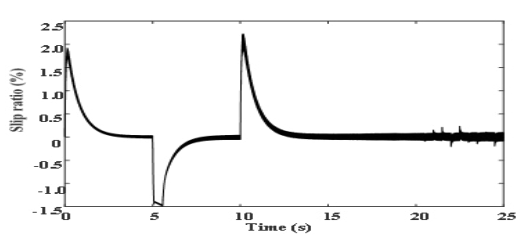

Figure 9: Slip ratio

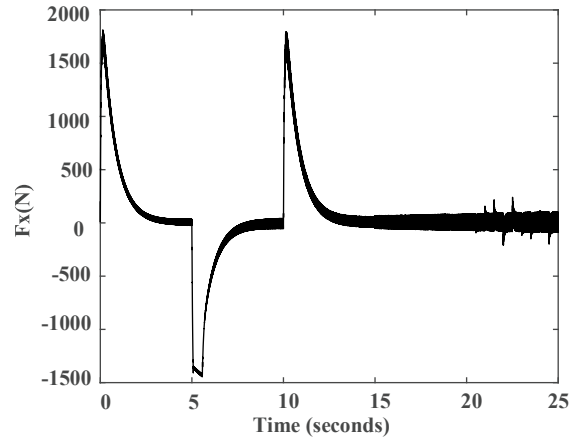

Figure 10: Longitudinal force F_x

5. CONCLUSION

In the current study, a mathematical model of BLDC motor and the tire based on Pacejka model was considered to be implemented in the model of the quarter of vehicle, then a fuzzy control approach was designed in order to follow the desired speed profile in an electric vehicle and at the same time prevent the wheels from slipping, moreover to enhance the response of the system in acceleration phase and deceleration phase in different condition of the road.
thus, the designed controller offers robustness and adaptability and increases the system performance in both dynamic and steady-state, resulting in a smooth driving experience, a safe driving experience, no error in the steady-state, and an acceptable anti-interference capability Transitional state.

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